

12.1 Overview

The blood, saliva, and stool biologic specimens collected from Cases and Controls at all three Ghana Breast Health Study sites will be shipped to the NCI Biorepository in Frederick, MD in the United States on a regularly scheduled basis.

To facilitate export of the specimens, the services of an NCI Contractor, PPD (Pharmaceutical Product Development, LLC), will coordinate and supervise specimen shipments from Ghana to the United States, and will provide detailed guidelines and directives to the sites for each shipment.

12.2 Regulatory Requirements

Individuals responsible for shipping biologic specimens by air transport must be trained in accordance with the criteria set forth by the International Air Transport Association (IATA) and published in the IATA Dangerous Goods Regulations each year. The IATA regulations govern the packing, marking and labeling of shipments of biologic specimens. The shipper, in this case PPD or any other company contracted to PPD for these services, is responsible to adhere to these regulations.

12.3 Shipping Plan for the Ghana Breast Health Study

12.3.1 Specimens to be Shipped to the United States

All saliva, blood, and stool specimens collected during the study will be shipped to the NCI Biorepository, with exception of the blood clot which will remain in Ghana. Any extra breast tissue cores, either as fixed tissue or in paraffin blocks, which are not needed for diagnosis will also be shipped to the NCI Biorepository.

12.3.2 Shipment Schedule

Shipments will be scheduled as determined by NCI and PPD in coordination with GBHS Staff. laboratories to the NCI Biorepository. Shipments will be scheduled to avoid known holidays in both countries or potential interruptions in government functioning (e.g. national elections in Ghana or a government shutdown in the United States) that could delay the proper delivery of specimens. Specific shipment dates will be arranged in advance between GBHS staff, NCI, and PPD.

12.3.3 Shipment Types: Ambient and Frozen

Blood products (serum, plasma, buffy coat and red blood cells) as well as stool samples will be shipped frozen in their freezer storage boxes. PPD will provide liquid nitrogen (LN2) shippers for the blood and stool samples (dry ice would also be an acceptable shipping method but availability is limited in Ghana). Saliva samples and tissue samples will be shipped at room temperature.

Shipping supplies for both frozen and ambient shipments will be provided by PPD. The site will be responsible for preparing the specimens for shipment. PPD or their affiliate will be responsible for packing the shipment.

12.4 Procedures for Preparing a Shipment to the United States

12.4.1 Pre-shipment Procedures

In preparation for making a shipment to the NCI repository:

- Westat will email GBHS staff a shipping manifest, which is a list of the specimens to be shipped by their type and ID. For blood and saliva, Westat or NCI will discuss with GBHS laboratory staff the number of boxes available to be shipped prior to preparing the manifest. For tissue samples, Westat or NCI will request a manifest from the pathologists in a format to be provided. On the tissue manifest, PIDs and accession numbers for the tissue samples will be required and will undergo QC review at Westat prior to shipment to ensure the link between specimens and subjects is accurate and complete.
- To the best of their abilities, GBHS staff should use the shipping manifest provided by Westat to select the specimens to be shipped to the United States. For shipment of samples at ambient temperature, GBHS staff should thoroughly crosscheck the samples with the shipping manifest and send only those samples on the list. For samples that

need to be shipped frozen, the specimens should be organized quickly for shipment in order to prevent any thawing of samples. NCI may ask Ghana staff to simply send a certain number of sample boxes by type, regardless of Sample ID, in order to prevent thawing of samples that would occur if the samples needed to be organized differently for a shipment. It is understood that dry ice is not commonly available to maintain the temperature of the samples during the organization process.

- The number of specimens included in the shipment **may not exceed** the number of specimens on the manifest. This is because the import permit allowing the specimens into the US is restricted to the number of specimens on the manifest.
- Study staff should assist with preparing any in-country paperwork requirements for shipment.

12.4.2 Shipment Procedures – Frozen Samples

- The courier will be another company hired by PPD. The courier is expected to notify the hospital with a minimum of 3 to 5 days pre-notification of when the sample pick-up time will occur.
- Two configuration of the shipment will depend upon the capacity of the LN2 dewars to be used for the shipment. Typically one LN2 dewar will hold five 2-inch tall cryoboxes with the dimensions of 5-inches(l) x 5-inches(w) x 2-inches (h). Each cryobox holds 81 – 2ml cryovials. If other dewars are used by the courier, NCI will notify the laboratory staff at the site as to the number of boxes to be placed in each dewar and possibly the sample types to be included in each dewar.
- The dewars should contain racks to hold the cryoboxes securely for transport. If the dewars arrive without racks, the laboratory staff should inform NCI immediately.
- At the time of actual shipping, there should never be liquid in the dewars. The dewars are what is referred to as ‘dry shippers’ and all LN2 is absorbed into the “honeycomb” lining of each dewar. If liquid is in the dewars when the specimen boxes are about to be inserted for shipment, the laboratory staff should inform NCI immediately.
- PPD will provide the airline with the list of samples contained in the shipment prior to the flight’s departure. A hardcopy shipping manifest will also accompany the dewar.

12.4.3 Shipment Procedures – Ambient Samples

- Saliva – Saliva samples will be shipped in the cardboard boxes provided by NCI that hold 49 samples each (7 rows x 7 columns). These boxes will be placed in a shipping container provided by PPD or another company hired by PPD. Prior to placing in the

shipping container, laboratory staff should place a rubber band around the saliva box to ensure that the lid stays attached during shipment. The rubber-banded saliva box should then be placed in a Biohazard Bag, which was provided by Westat with the supply shipment. An absorbent strip, also provided by Westat, should be placed in the Biohazard Bag as well, and then the bag should be sealed. Place as many saliva boxes into the shipping container as instructed by PPD or their affiliate.

- Tissue samples- Tissue cassette(s) should be transferred to a Seal-A-Meal® bag, and vacuum sealed with a Seal-A-Meal® vacuum unit. The sealed vacuum bag can be stored for up to six weeks at room temperature (not to exceed 40° C). Up to 12 cassettes can be packed and shipped per bag at ambient temperature. These bags will be placed in the ambient shipper at the time PPD or their affiliate arrive to collect the samples for shipment.
- Enclose a printed copy of the specimen manifest into the ambient shipping container.